

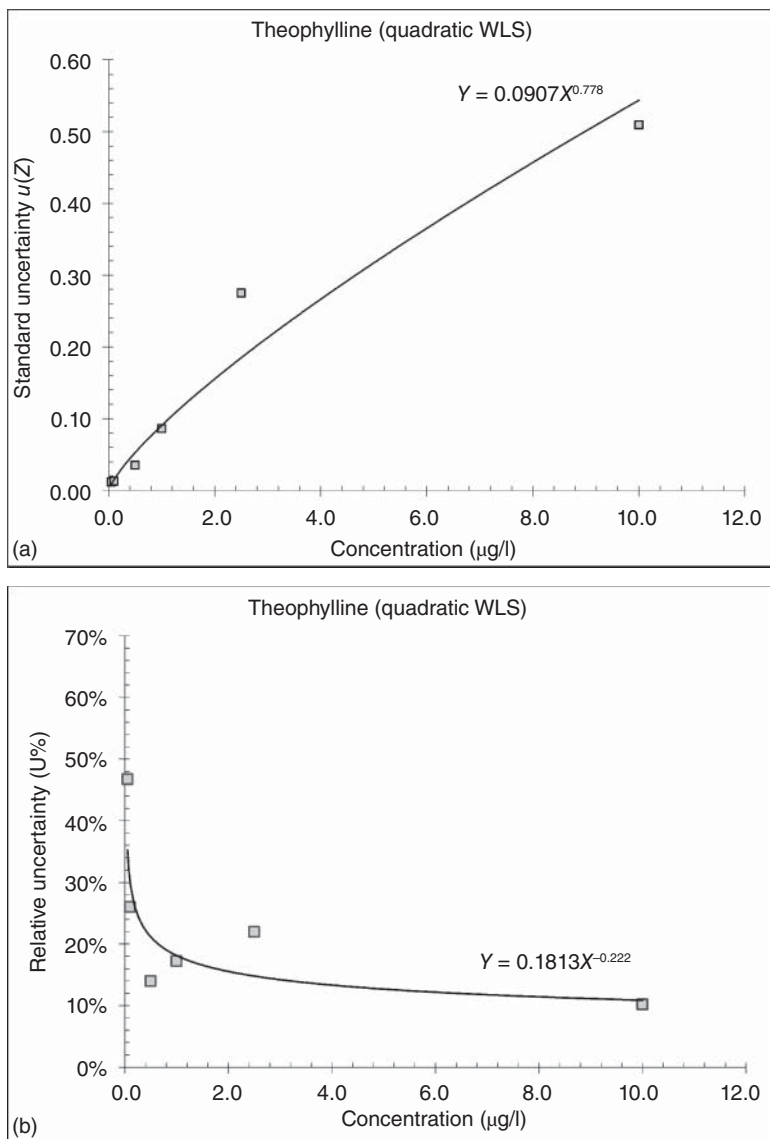


Figure 7.8 (a) THEOPHYLLINE – standard uncertainty function,
(b) THEOPHYLLINE – relative uncertainty function.

not necessarily true percentages as shown in Eqs. (7.30) and (7.31), and preliminary precautions must be taken. In Excel worksheets, the data display may not reflect the internal format. The same data worth 0.468, if the applied format is “Percentage” will be displayed as 46.8%, if it is “Number,” it is 0.468. The coefficients calculated with Resource M, especially the power coefficient, do not consider the display format. If the model is used routinely, it will be necessary to ensure that the data are input in the same format as used to estimate the coefficients of the function. The

parameters of the different uncertainty functions can be connected, depending on the display format, as follows:

	$u(Z)$	$UR \% (Z)$	$UR \% (Z) \times 100$
Constant:	$a = 0.0907$	$2a = 0.1813$	$2a = 0.1813$
Power:	$b = 0.7780$	$b - 1 = -0.2220$	$1 + b = 1.7780$

7.5.2.1 How to Interpret a Power Function?

From the previous results, the power coefficient b is positive regarding the standard uncertainty but negative for the relative uncertainty. While the linear function, such as the calibration straight line, is well known to analysts, the power function is less commonly considered. The latter is parametrized by two coefficients, noted a and b . To understand their respective roles, it is necessary to recall the classic properties of power notation:

$x^0 = 1$	$x^{-b} = \frac{1}{x^b}$	$x^{-1} = \frac{1}{x}$
$\sqrt{x} = x^{\frac{1}{2}} = x^{0.5}$	$\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} = x^{-0.5}$	$\frac{1}{\sqrt[3]{x}} = x^{-\frac{1}{3}} = x^{-0.333}$

Comparable to the Horwitz model, the power function is usually the most appropriate model to establish the relationship between MU parameters and concentration. The power coefficient b for the relative uncertainty function is always < 0 giving a concave shape to the curve. This can be explained by the decrease of $UR \% (Z)$ when the concentration Z is increasing. The curve concavity underlines this property.

Figure 7.9 illustrates various power functions when coefficient b varies between $[0, -0.6]$ and coefficient $a = 0.2$ remains constant. These values are classically encountered, for instance, if $b = -\frac{1}{2}$ it means that $UR\% = \frac{1}{\sqrt{Z}}$. The more negative b is, the more pronounced the incurvature, and the more b approaches 0, the more the function flattens and becomes a horizontal line, indicating that the relative uncertainty is constant and equal to a . The constant a of the function expressed in % represents the relative uncertainty when the concentration is 1 unit. In this figure, as $a = 0.2$ the relative uncertainty is always 20% when $Z = 1$. When considering the standard uncertainty, the curve is convex. This illustrates the increase of the $u(Z)$ with concentration. The higher the coefficient b , the faster the increase. In practice, typical b values are between 0 and 2.0. For example, $b = 0.845$ in the Horwitz model.

7.6 Concept of Coverage Interval

7.6.1 Origin of Coverage Interval

Although the initial concept of MU can be dated back to 1977, it was not until the early 2010s that a practical document was published under the reference ISO/IEC Guide 98-4 on *The Role of Measurement Uncertainty in Conformity Assessment*

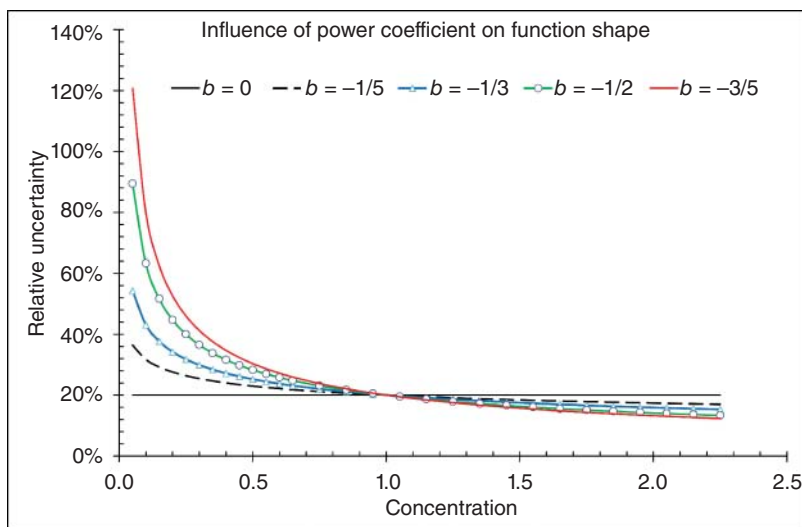


Figure 7.9 Different power functions when power coefficient b varies, and $a = 0.2$ remains constant. Concentration in arbitrary units.

to explain how MU can be used in decision-making. Two other paradigm shifts occurred during this period of slow maturation.

- The notion of a single true value for a measurand has been discarded in favor of the coverage interval.
- The definition of accuracy has been modified, making the use of this term for the MAP inconsistent with its metrological definition.

Of course, the notion of a single true value, introduced by statisticians many years ago, has practical problems. Therefore, the concept of MU implies that, for a given analyte and a given result, there is not a single true measurement value but an infinite number of possible true values scattered around the observed result. These possible true values are not equal and can be attributed to the analyte with varying degrees of reliability or probability. However, they are all compatible with the observations and with the knowledge that the analyst may have about the analyte.

This explains why the term “true value” has disappeared from GUM in the 2012 release, whereas it was present in previous versions. Instead, an analytical result is expressed as a coverage interval, defined as “the set of true values of a measurand with a specified probability based on the available information.” This definition is related to the notion of statistical dispersion interval, as explained in Section 5.3.

In view of the importance of this work on MU for the industry, in 1997, the BIPM set up a specific committee to prepare and draft documents relating to the estimation of MU. This is the Joint Committee for Guides in Metrology (JCGM) which is responsible for the Guide to the Expression of Uncertainty (GUM) and the VIM. For instance, next to developing these concepts, proposals such as replacing the *TAE* with MU in expressing the results from medical laboratories have been published [31].

In practice, there is no contradiction between these two parameters. They are not mutually exclusive because they do not consider the same objects of study. The *TAE* qualifies the method's performance, while the *MU* seeks to qualify the measurement itself [32].

Given the ultimate goal of the analytical process, which is to provide a result that allows an informed decision, *MU* is therefore preferable. The use of *MU* as a single validation criterion for quantitative methods could be considered relevant if a universally accepted estimation procedure could emerge for the analytical sciences, which is not yet the case, as explained in Section 7.1.

MU was originally adopted for physical measurements but has only recently become familiar to chemical analysts through the ISO 17025 standard for laboratory accreditation. Uncertainty, "a parameter that characterizes the dispersion of values attributed to a measurand, based on the information used," appears more than 30 times in the 2005 version of ISO 17025 and with equal frequency in the 2017 version, while it was absent in previous versions.

Estimating *MU* consists in evaluating the dispersion of all the measurement values that could be obtained from a given sample. With a quick reading of this definition, one might be tempted to propose, for an estimation procedure, the interval of dispersion of measurements directly derived from the distribution function of a random variable. For example, if measurements Z_i are exactly distributed according to a Normal distribution of theoretical mean μ and variance σ^2 , denoted $\mathcal{N}(Z, \sigma^2)$, it is possible to confirm that 95% of the values are distributed in the following interval:

Dispersion interval

$$[\mu - 1.96 \times \sigma, \mu + 1.96 \times \sigma]$$

This approach is a misinterpretation of the definition of *MU* because such an interval assumes that μ and σ are known and does not consider all the "information used" to obtain the measurement values. For instance, the variability of the conditions under which the measurements are made nor the sources of error, such as measurement bias, are not considered.

To circumvent this problem, it is possible to obtain a prediction interval based on estimated values of μ and σ . The prediction interval is discussed in more detail Section 5.3. Moreover, from an economic point of view, the most serious drawback is it would require systematic replication of measures to estimate coverage interval; the prediction interval estimated on one day or one sample could not be transferable to another as measuring conditions may have changed.

Given the relatively high cost of many analyses and the intense pressure on prices, laboratories would find it impractical to provide this parameter. Therefore, a more comprehensive approach to estimating *MU* that is transferable from one measurement to another or one sample to another is needed. To reach this goal, it must be remembered that *MU* is a parameter that characterizes the dispersion of measurements "once all causes of variability are taken into account." The procedure proposed by the GUM for estimating *MU* consists, for each measurand, in making an inventory as exhaustive as possible of all the components contributing to the

variability, as explained in Section 6.2. At this point, three prominent issues should be emphasized:

- (1) The requirement in the ISO 17025 standard:

“5.4.7.2 Testing laboratories shall also have and apply procedures for estimating measurement uncertainty, except where test methods preclude such rigorous calculations. In some cases, it is not possible to make a metrologically and statistically valid estimate of measurement uncertainty. In such cases, the laboratory should at least attempt to identify all components of uncertainty and make the best estimate possible, while ensuring that the manner of reporting does not give an exaggerated impression of accuracy”.

- (2) MU characterizes a measurement or a result and not a method. It contrasts with several classic validation parameters, such as the coefficient of variation of repeatability or the recovery yield, even if these parameters participate in the MU.
- (3) When estimating MU, “all biases are assumed to be ignored or corrected for by appropriate and validated factors.” In this case, the uncertainty of the correction factors must be included in the final evaluation, as discussed in Section 8.4.2.

In Section 7.2, a global solution for estimating MU for analytical sciences is described. It is global in that it considers all sources of uncertainty simultaneously. Such an approach is sometimes referred to as “holistic” [33]. In Section 4.1.2, the evolution of the ambiguous concept of single true value leading to the preferential use of the coverage interval to express a measurement value is explained.

Figure 7.10 shows the different notions used to define the coverage interval. The coverage factor is fixed here at 2, which implies that the *coverage interval* contains about 95% of the possible true values of the measurand.

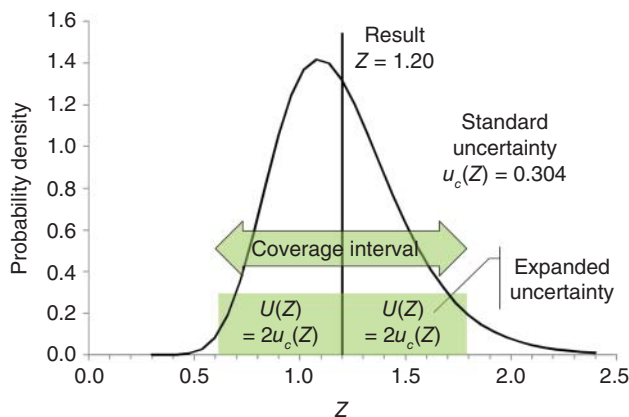


Figure 7.10 Coverage interval and measurement uncertainty.